NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2019-2020

MH1812 - DISCRETE MATHEMATICS

December, 2019 TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **SEVEN (7)** questions and comprises **FOUR (4)** printed pages.

2. Answer **ALL** questions. The marks for each question are indicated at the end of each question.

3. Answer each question beginning on a **FRESH** page of the answer book.

4. This **IS NOT** an **OPEN BOOK** exam.

5. Candidates may use calculators. However, they should write down systematically the steps in the workings.

QUESTION 1. **(10 marks)**

Decide whether the following argument is valid:

$$T \to (E \vee M);$$
$$S \to \neg E;$$
$$T \wedge S;$$
$$\therefore M.$$

QUESTION 2.

(a) Find the solution of the recurrence relation $a_n = 2a_{n-1} + 1$ with $a_1 = 1$.

(10 marks)

(b) For all $n \geq 1$, prove the following by mathematical induction:

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \cdots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n}.$$

(10 marks)

QUESTION 3.
In how many ways can a committee of 5 be formed from a group of 11 people consisting of 4 teachers and 7 students if

(a) there is no restriction in the selection? **(5 marks)**

(b) the committee must include exactly 2 teachers? **(5 marks)**

(c) the committee must include at least 3 teachers? **(5 marks)**

(d) a particular teacher and a particular student cannot be both in the committee? **(5 marks)**

QUESTION 4. **(12 marks)**
Prove for three sets A, B, and C, if $A \cap C = B \cap C$ and $A \cup C = B \cup C$, then $A = B$.

QUESTION 5. **(16 marks)**
The relation R is defined on the set of integers $\mathbb{Z}$ as follows. For all $x, y \in \mathbb{Z}$,

$$x \; R \; y \iff 3 \mid (x^2 - y^2).$$

Determine if R is an equivalence relation, and if so, show the equivalence classes.

QUESTION 6.
Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, and $f : A \to B$.

(a) How many such functions f are there ? **(4 marks)**

(b) How many such onto functions f are there ? **(6 marks)**

(c) How many such one-to-one functions f are there ? **(4 marks)**

QUESTION 7. **(8 marks)**
In Sam's messy dresser drawer, there is a jumble of 6 red socks, 7 blue socks, 9 green socks, and 5 yellow socks. If Sam grabs a handful of socks without looking at what he's taking, what is the minimum number of socks Sam has to grab in order to guarantee that he has at least 4 socks of the same color?

END OF PAPER